

Intraday Signal Platform: Implementation Plan (IDesign)

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Scope and locked decisions

The platform is a long-only, human-in-the-loop signal system: it scans about 600 liquid US stocks, estimates each setup's probability of reaching its target before its stop, and shows the best few to trade by hand on XTB. Every design choice serves one test: are the stated probabilities honest after XTB's costs and a 30 to 90 second reaction delay?

Area	Decision	Reason
Execution	Manual on XTB. No broker API anywhere in the system.	XTB shut its API on 14 March 2025. Manual execution removes order management, kill-switches and most of the original code risk.
Direction	Long only	Real shares cannot be shorted. Halves the modeling work.
Instruments	Real US shares in a USD account	Avoids XTB's 0.5% conversion fee on each leg (about 1% per round trip). Commission-free up to €100,000 monthly turnover, then 0.2% (minimum €10).
Position size	About \$500 notional, sized from a fixed dollar risk of about \$5	Keeps the validation phase cheap while still using real money and real hesitation.
Universe	About 600 stocks: price above \$10, 20-day dollar volume above \$50M, ATR above 2% of price, spread under 0.10%, plus a daily in-play list	Liquid names give honest, repeatable statistics. Small caps are a separate later experiment.

Area	Decision	Reason
Data	Alpaca free tier (history since 2016 excluding the latest 15 minutes, live IEX), Alpaca news (Benzinga content), SEC EDGAR	\$0 for research. Paid real-time data is decided by experiment E1, not assumed.
Catalyst classifier	Jev (TypeSafe) behind an interface, kept only if it wins the bake-off (E9)	Early access, no independent benchmarks yet.
Compute	Your own computer. Python, Parquet, DuckDB. No Redis, no cloud.	One user, one process, human in the loop.
Success test	Calibrated probability and positive expectancy after costs and delay, out of sample	A win rate alone says nothing without the reward-to-risk ratio.

Out of scope: automated execution, short selling, small caps, tick-level order-book research, and any promise of a guaranteed edge.

What changed from the original spec

The original document is kept for its two-tier idea (cheap technical screening first, language model only on the shortlist) and its layered structure. Most of the rest is replaced, because it was wrong, out of date, or untested.

Original element	Problem	Replacement
XTB broker accessor	XTB API access was discontinued on 14 March 2025	Removed. Manual execution on XTB.
Trading 212 and Alpaca accessors, execution manager, kill-switch, position lifecycle	Not needed for manual trading. The Trading 212 API supports only Invest and ISA cash accounts, so it could not execute short signals.	Removed. An outcome tracker simulates what each signal would have done.

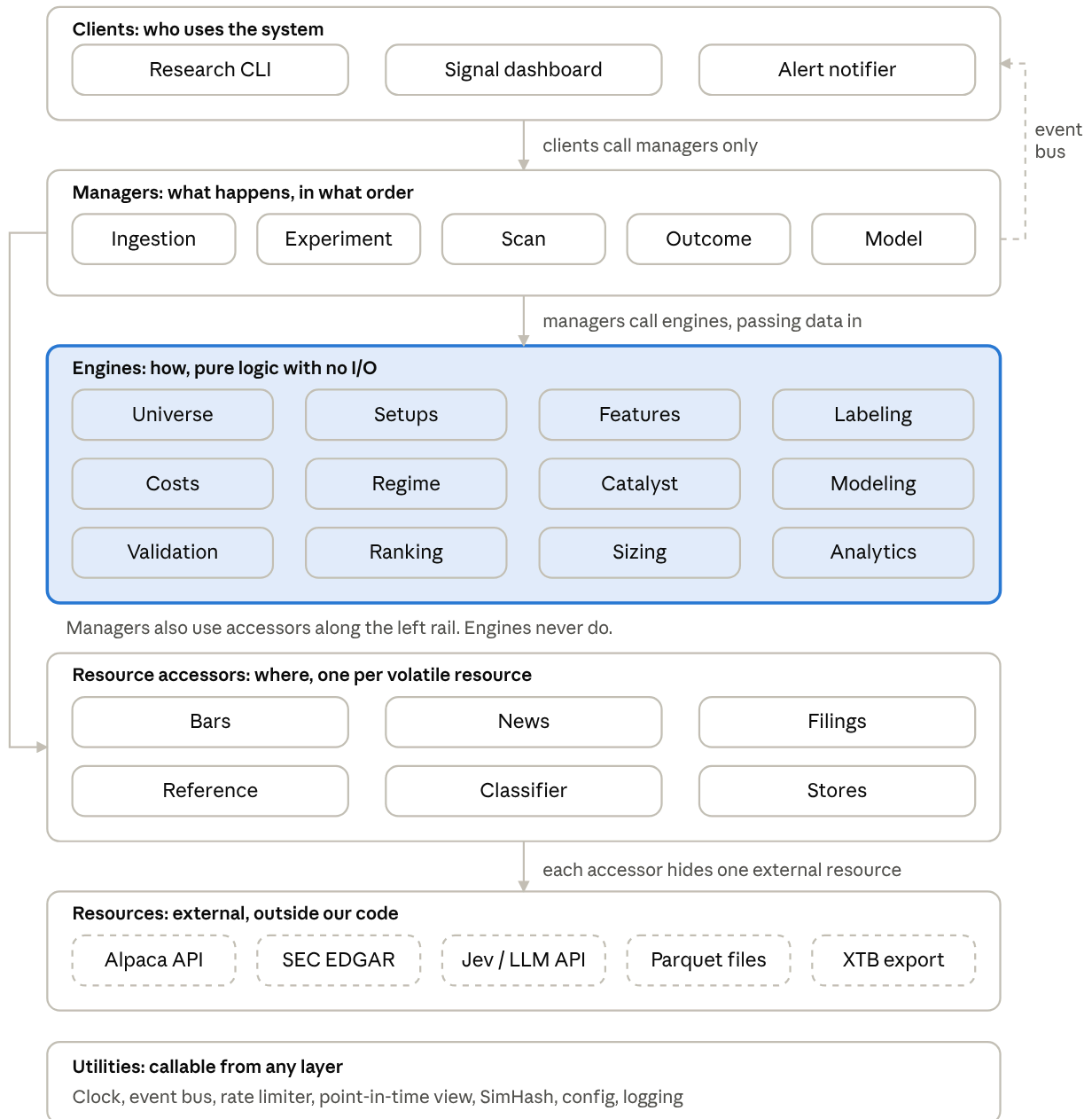
Original element	Problem	Replacement
Alpaca day-trade check (equity below \$25,000 and daytrade_count at 3 or more)	The pattern day trader rule was eliminated on 4 June 2026 and Alpaca removed the fields	Removed with execution
Composite score with weights 0.35, 0.35, 0.15, 0.15 and a cutoff of 70	Weights are unvalidated. Without news the maximum score is 65, so no technical-only setup could ever pass.	Calibrated probability from a meta-labeling model, ranked by expected value in R (multiples of risk)
Language model proposes entry, stop and targets	Prices should come from the chart, not from text. The "conviction score" is not a calibrated probability.	Prices set by deterministic code. The classifier only labels the catalyst.
Strategy B: 2σ in the text, 1.5σ in the formula, one bar at $t-2$	Self-contradictory and too narrow to find real reclaims	Parameters (band width, lookback window) tested in E4
Engines call the language model	A network call belongs in a Resource Accessor under IDesign	Classifier sits in an Accessor. Engines stay pure.
Manager pushes straight to the portal	A Manager calling up into a Client breaks the layering	Publish and subscribe through an in-process event bus
Redis, WebSocket fan-out, 25 ms latency targets	Built for automation and many users. Signals live 45 to 114 seconds and wait for a human click.	One process, Parquet and DuckDB. Latency is measured in seconds, not milliseconds.
No backtest, no validation, no market data source	The part that decides whether any of it makes money	Experiments E0 to E14 below

The document's citations were mostly book summaries, forum posts and blogs, so treat its claims as unverified until an experiment or a primary source confirms them.

IDesign decomposition

The system is split by what is likely to change, not by what it does, following Juval Löwy's IDesign method. Vendors, strategy rules, classifiers and the user interface change for different reasons, so each sits behind its own boundary. The same engines serve backtests and live scans.

Clients call managers, managers use engines and accessors, nothing calls up



architecture · 5 layers plus utilities

Engines hold every piece of logic an experiment changes, and the only upward path is the event bus that carries signals to the dashboard and the notifier.

Component catalog

Component	Layer	What it hides	Used in
Ingestion	Manager	Downloads, backfills, daily updates and data checks	E0
Experiment	Manager	The sequence of a run: build dataset, label, fit, validate, report	E1 to E12
Scan	Manager	One scanning cycle, in replay or live, publishing signals to the event bus	E13
Outcome	Manager	Tracking each signal's result and your trade journal, live calibration	E13, E14
Model	Manager	Retraining, promotion and rollback of model versions	E7, E13
Universe	Engine	Point-in-time liquidity and tradability filters	E0
Setups	Engine	Strategies A, B and C as configurable detectors	E3, E4
Features	Engine	Feature definitions from bars, context and catalyst labels	E7, E8
Labeling	Engine	Triple-barrier outcomes with entry delay and costs	E3, E5, E6
Costs	Engine	XTB spread, slippage and fee model	E2
Regime	Engine	Market state from SPY, QQQ, VIX and breadth	E11
Catalyst	Engine	Mapping classifier probabilities and filing items to features and vetoes	E9, E10
Modeling	Engine	Training, calibration and prediction	E7
Validation	Engine	Purged walk-forward, deflated Sharpe, holdout rules	All
Ranking	Engine	Expected value in R, daily top picks, concentration flags	E12

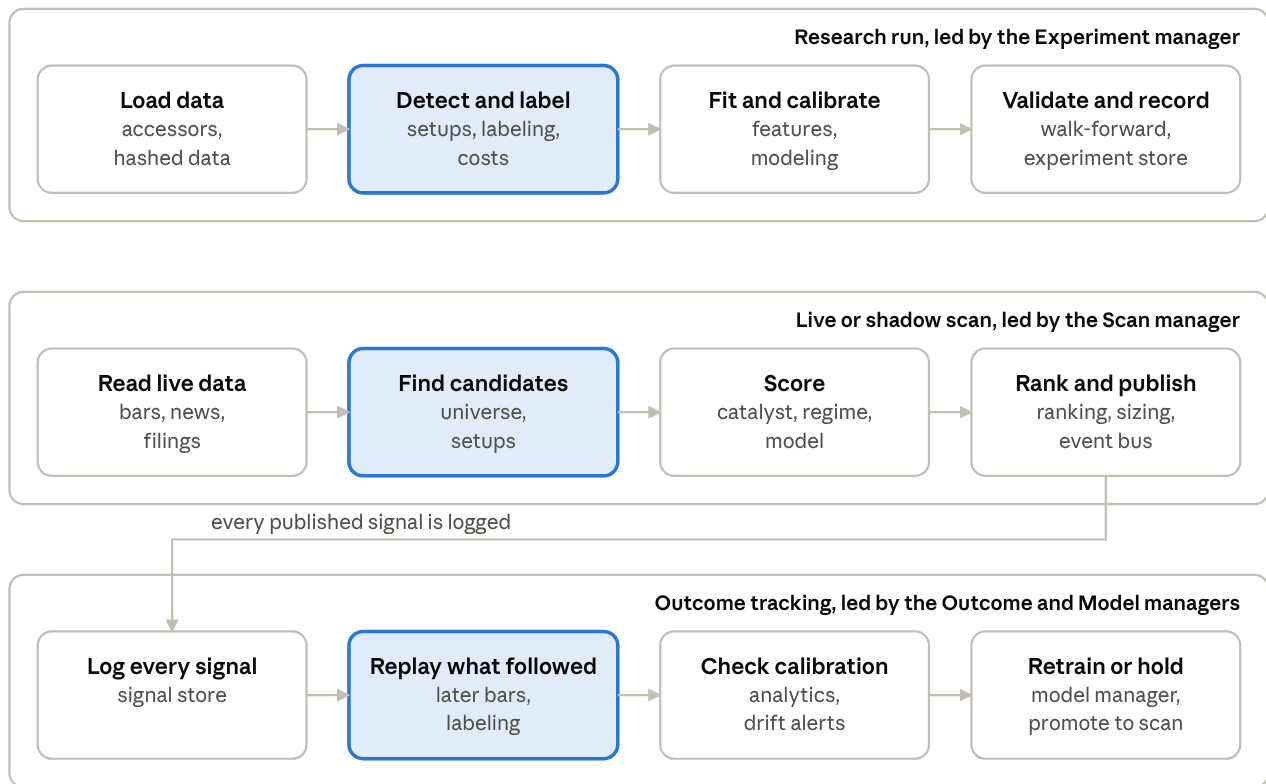
Component	Layer	What it hides	Used in
Sizing	Engine	Fixed-dollar-risk size, notional cap, turnover tracker	E12, E14
Analytics	Engine	Calibration curves, drift detection, attribution	E13
Bars	Accessor	Alpaca versus local replay, feeds, adjustments, rate limits	E0, E1
News	Accessor	Alpaca news format and paging	E9, E10
Filings	Accessor	EDGAR formats, rate limits, item codes	E9, E10
Reference	Accessor	Asset list and the XTB tradable list	E0
Classifier	Accessor	Which model labels the text, its prompts and output schema	E9
Stores	Accessor	File layout for signals, experiments and models	All

Call rules and runtime flows

Six rules keep the layers honest, and the three runtime flows below show them at work. A rule that cannot be checked by a test is not a rule, so each of these is enforced by the architecture tests.

1. Clients call managers only, never engines or accessors.
2. Managers call engines and accessors. They never call other managers directly and coordinate through the event bus instead.
3. Engines call nothing: no accessors, no managers, no other engines. This is stricter than IDesign, which lets engines use accessors, and it makes engines testable with plain inputs.
4. Accessors never call each other, engines or managers.
5. Nothing calls upward. Signals reach the dashboard and notifier through the event bus.
6. Utilities may be used anywhere but hold no business logic. Time and data enter explicitly: every engine call receives an as_of time and a point-in-time view, and nothing reads the clock on its own.

One set of engines serves research, the live scan and outcome checks



☐ Highlighted steps run the same engine code, so research results carry over to live

runtime flows · 3 lanes

The research run and the live scan differ only in the accessor (stored Parquet files versus the live API) and the clock. Experiment E13 tests this by replaying live days through the research path and requiring the same signals.

Data architecture

All data lives as immutable Parquet files on your disk, queried with DuckDB, and every dataset is identified by a content hash so any experiment can name the exact data it used.

Dataset	Source	Grain	Location under data/	Notes
Asset reference	Alpaca assets, active and inactive	symbol	reference/assets.parquet	Exchange and name drive the exclusion of ETFs, funds and warrants.
Daily bars	Alpaca, SIP feed	symbol, day	bars_1d/year=YYYY.parquet	Split-adjusted. Feeds the point-in-time universe and ATR baselines.
Minute bars	Alpaca, SIP and IEX	symbol, minute	bars_1m/feed=F/month=YYYY-MM/	Extended hours kept. Fetched only for months a symbol was in the universe.
News	Alpaca news (Benzinga content)	article	news/month=YYYY-MM.parquet	Keeps created_at, updated_at, headline, summary, symbols.
Filings	SEC EDGAR submissions	filing	edgar/filings.parquet	8-K item codes, offering forms (S-1, S-3, 424B), acceptance time.

Dataset	Source	Grain	Location under data/	Notes
Catalyst labels	Classifier accessor	article, model version	catalyst/	Class probabilities, keyed by model version.
Setups and labels	Engines	candidate	research/setups, research/labels	Triple-barrier outcome with delay and costs applied.
Signals and outcomes	Scan and outcome managers	signal	live/signals, live/outcomes	Every signal is logged whether or not you trade it.
Trade journal	You	trade	live/journal	Used to measure execution shortfall.
Experiment runs	Experiment store	run	experiments/	Config, code commit, dataset hashes, metrics.

Point-in-time rules

1. Every record carries the time it became knowable (`available_at`), not only the event time. A bar is available at its close. A news item is available at `created_at` plus a lag, set to 60 seconds until E10 measures the real delay. A filing is available at its SEC acceptance time.
2. Universe membership for day t uses data through the close of day $t-1$ only.
3. Every engine takes an `as_of` timestamp and sees only records with `available_at` at or before it. A test poisons all later data and checks the output does not change.
4. Prices are split-adjusted but not dividend-adjusted, so levels match what you see on XTB.

5. Timestamps are stored in UTC and sessions are computed in America/New_York. The US and Europe change clocks on different dates, so Romanian offsets are never hard-coded.
6. Raw downloads are never edited. Fixes happen in derived layers.
7. Free-tier SIP queries must end at least 15 minutes in the past, so the downloader clamps its end time. The free plan also allows 200 historical calls a minute and only 30 live WebSocket symbols, so live free data means polling 1-minute bars over REST, not streaming.
8. Survivorship bias is handled by building the universe from the asset list including inactive symbols and auditing coverage in E0. Alpaca's list may not contain every delisted ticker, which E0 must measure rather than assume.

Volume and time estimates

These are approximate and must be replaced by measurements from the pilot (200 symbols, 3 years). Daily bars for all US stocks over 10 years are about 20 million rows and a few minutes of downloading. Minute bars for 600 symbols over 7 years with extended hours are several hundred million rows. At 10,000 bars per request and 200 requests per minute, expect roughly 5 hours of downloading per feed and on the order of 10 GB of compressed Parquet. That revises my earlier "a few gigabytes" figure upward, so budget a 500 GB drive and run the pilot before the full backfill.

Experiment framework

Fifteen experiments (E0 to E14) decide whether each next layer gets built, and each has a written gate. Failing a gate means redesign or stop, never a looser threshold. The gate values below are my proposals and should be confirmed before the first run.

R is profit divided by the initial risk (entry minus stop). Every result is reported in R, after XTB costs and your entry delay, and the ranking metric is expected value per trade (ignoring time-outs):

$$EV_R = p \cdot R_{\text{target}} - (1 - p) \cdot c$$

Here p is the probability the target is hit before the stop, R_{target} is the reward-to-risk ratio and c is the round-trip cost expressed in R.

Rules for all experiments:

- Each run is one config file, one code commit, the dataset hashes and its metrics, saved by the experiment store.
- Every parameter combination tried is counted. The count feeds the deflated Sharpe ratio, which discounts results for how many things were tried.
- Training and tuning use purged walk-forward splits. The last 12 months are a locked holdout, evaluated once at gate G2.
- A fast failure is a good outcome. E3 in particular can end the project cheaply.

Data and costs (E0 to E2)

ID	Question	Method	Gate (proposed)
E0	Is the history trustworthy and free of survivorship bias?	Check 50 stocks delisted after 2016 against the downloaded data. Count missing regular-session minutes for universe names. Scan for price jumps above 40% with no split. Compare daily bars with daily totals rebuilt from minute bars.	At least 90% of sampled delisted names present. Missing regular-session minutes under 0.5%. No unexplained jumps. Daily volume within 2% of the minute total.
E1	Can free IEX live data replace paid real-time data?	On stored IEX and SIP minutes, scale IEX volume by each stock's trailing ratio and compare RVOL and VWAP with SIP at 09:35, 09:45, 10:00 and 10:30 New York time. Live free data is polled over REST because the free WebSocket covers only 30 symbols.	Median RVOL error under 15% by 10:00, and the RVOL at least 2.5 flag agrees with SIP in 90% of cases at 09:45. A fail means: pay the \$99 plan, start signals later, or use price-only features at the open.

ID	Question	Method	Gate (proposed)
E2	What do XTB fills really cost?	Log XTB quotes next to the Alpaca SIP midpoint for 12 stocks, three times a day, for 5 days. Add a volume-based slippage model and the 0.2% commission above the monthly limit.	Modeled spread plus slippage within 30% of observed. Output is the cost model every later experiment uses.

Signal research (E3 to E8)

ID	Question	Method	Gate (proposed)
E3	Does any rule-based setup have an edge before any machine learning?	Run strategies A, B and C alone. Enter 60 seconds after the signal at the then-current price plus slippage. Score by triple-barrier outcome with costs, walk-forward by year.	Positive mean R after costs, pooled out of sample and in at least 60% of years, with at least 300 signals. If no strategy passes, stop and redesign before building more.
E4	Is the edge a knife-edge of tuned thresholds?	Grid around each threshold: RVOL 2.0 to 3.5, VWAP band 1.0σ to 2.0σ , lookback window, ATR ratio. Plot the expectancy surface and record the trial count.	Neighboring settings keep at least half the best expectancy. Deflated Sharpe ratio stays positive after the trial count.
E5	Which stop, target and time limit work best?	Compare stops at $1\times\text{ATR}$ and at structure (below VWAP or the opening range), targets of 1.5R, 2R and 3R, time limits of 30, 60 and 90 minutes, chosen by nested validation.	The same choice wins in most years and beats the others by more than the standard error.

ID	Question	Method	Gate (proposed)
E6	Does the edge survive your reaction time?	Re-run E3 with entry delays of 0, 30, 60, 90 and 180 seconds.	Positive mean R at 90 seconds. The delay where half the edge is gone sets each signal's expiry and its do-not-chase price.
E7	Does a model filtering the rule candidates beat the rules alone?	LightGBM on setup, context, liquidity and time-of-day features, with purged walk-forward validation and isotonic calibration. Catalyst features join in E10.	Out of sample: calibration slope 0.8 to 1.2, Brier score better than the base rate, and the top third by predicted probability beats the unfiltered mean R by at least 0.1R.
E8	Is the model result real?	Shuffle labels (expect AUC near 0.5). Shift features one bar forward (expect a collapse). Drop each feature group. Run permutation importance.	Shuffled-label AUC between 0.48 and 0.52. Shifted features visibly degrade results. Feature groups that add nothing are removed.

Catalyst, regime and ranking (E9 to E12)

ID	Question	Method	Gate (proposed)
E9	Which classifier labels catalysts best?	Hand-label 500 to 1,000 headlines and filings (earnings, guidance, FDA, contract, offering or dilution, legal, analyst action, noise, plus new versus repeated). Compare Jev, a small language model, a fine-tuned ModernBERT or SetFit model, and rules on EDGAR item codes. Compare accuracy before and after each model's release date to detect memorized outcomes.	Offering and dilution recall at least 0.90 at precision at least 0.80. Macro F1 at least 0.80. Calibration error under 0.05. Before-and-after accuracy gap under 5 points.
E10	Do catalyst features improve signals, and is the news fast enough?	Add classifier outputs to the E7 model. Measure the delay from news timestamp to the first price move. Count the losing trades a dilution veto would have avoided.	Out-of-sample uplift in the top third's mean R beyond noise. If none, drop the language-model layer and keep only the EDGAR dilution veto. Also measure coverage: Alpaca's historical feed averages about 130 articles a day, so catalysts it misses count against it. Output is the real news lag.
E11	Should signals be suppressed on weak market days?	Add SPY, QQQ, VIX and breadth features. Compare with and without a flag when SPY is below its VWAP and falling.	Smaller worst-day and worst-week losses while losing no more than 10% of total expectancy.

ID	Question	Method	Gate (proposed)
E12	Does ranking by expected value beat simpler ranking, and does risk-based sizing help?	Compare the top 3, 5 and 10 per day ranked by expected R, raw probability, RVOL and at random. Compare fixed \$5 risk with fixed \$500 notional. Test sector-concentration flags.	Top 5 by expected R beats a random top 5 by at least 0.1R per trade. Risk-based sizing lowers result variance.

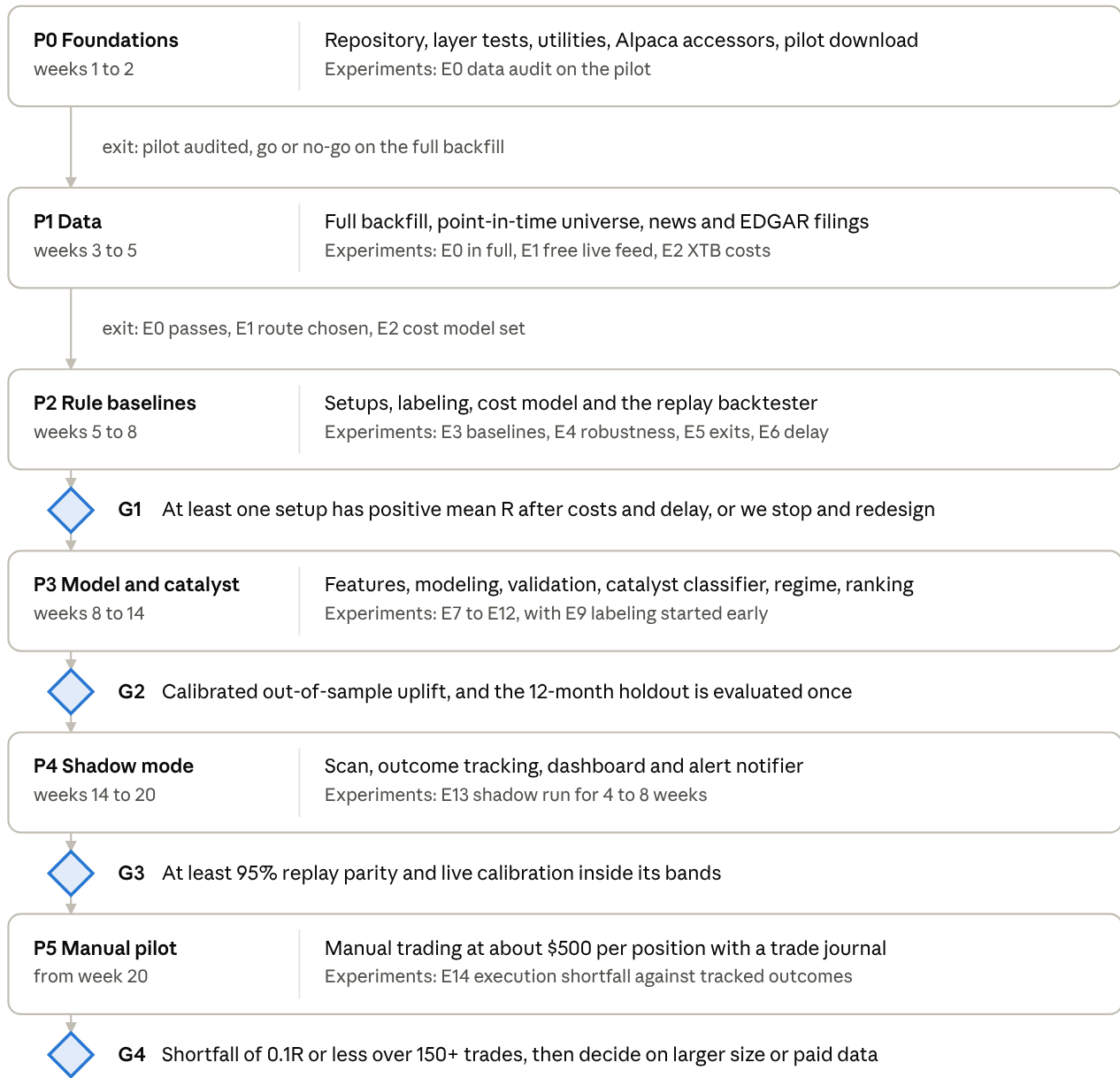
Live validation (E13 and E14)

ID	Question	Method	Gate (proposed)
E13	Does the live system behave like the backtest?	Run the scanner for 4 to 8 weeks without trading. Replay the same days through the backtest and compare signal lists. Track live calibration per probability bucket.	At least 95% identical signals on replay. Live calibration inside its confidence bands. No data outage over 5 minutes goes unnoticed.
E14	Do your own fills match the tracked outcomes?	Trade signals by hand at about \$500 and log every fill. Compare your R with the signal's tracked R.	After at least 150 trades, average execution shortfall of 0.1R or less per trade. Only then consider larger size or paid data.

Phased roadmap

The plan reaches a first manual trade in about 20 weeks, and four gates can stop it earlier at little cost. Week counts are my estimate for 10 to 15 hours a week and are not to scale in the picture.

Four gates can stop the project before any live trading



roadmap · 6 phases, 4 gates

The first two gates carry the most weight. G1 is the cheapest place to learn that the setups have no edge, and G2 is where overfitting would show up, because the holdout has not been touched until then. Phases P0 and P1 produce no trading signal at all, so a late start there delays everything: keep the pilot small and the audit honest.

Repository structure and contracts

The repository mirrors the IDesign layers one folder each, so a file's location tells you what it may import. The same engine code runs in backtest replay and in the live scanner. Only the accessor and the clock change.

```

signal-platform/
  pyproject.toml
  config/
    universe.yaml  strategies.yaml  costs.yaml
    experiments/E03_rule_baselines.yaml ...
  src/signalplat/
    contracts/      DTOs and Protocol interfaces only, no logic
    clients/        cli.py, dashboard/, notifier.py
    managers/       ingestion.py, experiment.py, scan.py, outcome.py,
model.py
  engines/         universe.py, setups.py, features.py, labeling.py,
costs.py,
                  regime.py, catalyst.py, modeling.py, validation.py,
                  ranking.py, sizing.py, analytics.py
  accessors/       bars_alpaca.py, bars_parquet.py, news_alpaca.py,
                  filings_edgar.py, reference_alpaca.py,
                  classifier_jev.py, classifier_llm.py,
                  signal_store.py, experiment_store.py, model_store.py
  utilities/       clock.py, eventbus.py, ratelimit.py, simhash.py,
                  pit.py, config.py, logging.py, ids.py
  tests/           unit/ integration/ architecture/ leakage/
  data/            (git-ignored)
  notebooks/

```

The interfaces below are the seams where experiments swap parts. Anything volatile (a vendor, a classifier, a strategy rule) sits behind one of them.

```

@dataclass(frozen=True)
class Candidate:
    symbol: str
    strategy: str          # "A_breakout" | "B_vwap_reclaim" |
"C_catalyst_gap"
    as_of: datetime        # UTC, when the setup became knowable
    ref_price: float
    stop: float
    targets: tuple[float, ...]

```

```

class BarsAccessor(Protocol):
    def minute_bars(self, symbols: Sequence[str], start: datetime,
                    end: datetime, feed: Feed) -> pd.DataFrame: ...

class CatalystClassifier(Protocol):
    def classify(self, items: Sequence[NewsItem],
                spec: ClassifierSpec) -> list[ClassifierResult]: ...

class SetupEngine(Protocol):
    def detect(self, view: PointInView, ctx: MarketContext,
               as_of: datetime) -> list[Candidate]: ...

```

Conventions:

- Python 3.12 with type hints, frozen dataclasses, no global state. Pandas backed by Arrow for data frames, DuckDB for queries over Parquet.
- Engines do no I/O, never read the clock and take a seed for any randomness. Where data is needed, the manager fetches it through an accessor and passes it in. This is stricter than IDesign, which lets engines call accessors, and it keeps every engine testable with plain inputs.
- Accessors return contract types, never vendor shapes, so replacing Alpaca changes one file.
- Managers sequence work and hold use-case state. They contain no formulas.
- Thresholds and weights live in YAML config, not code, so an experiment is a config change.
- One command-line tool, with a subcommand per manager, is the only client needed until the dashboard phase.
- Layer rules are enforced in continuous integration with import-linter, so an illegal import fails the build.

Testing and quality gates

The main risk in this project is not a crash but a result that looks good because of leakage, overfitting or bad data. The test suite is built around those three failures first.

Test type	What it proves	Where
Future-poisoning	Replace all data after as_of with garbage. Every engine's output must be identical. This catches lookahead.	tests/leakage
Replay parity	The live scan path and the backtest replay path produce the same signals for the same day.	tests/integration
Shuffled-label and shifted-feature	Model skill vanishes when labels are shuffled and collapses when features are shifted ahead.	tests/leakage
Golden values	VWAP, ATR, RVOL and the triple-barrier labeler match hand-calculated examples.	tests/unit
Property tests	Engines never return a stop above entry, a negative size, or a signal outside its universe.	tests/unit
Recorded-HTTP contract tests	Each accessor parses saved vendor responses into contract types, including error and empty cases.	tests/integration
Data quality checks	No duplicate (symbol, timestamp), no bars outside session hours, no unexplained jumps, volume totals consistent.	run after every ingestion
Architecture rules	Clients import only managers, managers import engines and accessors, engines import neither.	tests/architecture, import-linter
Reproducibility	The same config and dataset hashes give the same metrics.	experiment store

Statistical gates, applied before any result is believed:

- All numbers are out of sample, after costs and entry delay.

- Trial counts are logged and the deflated Sharpe ratio is reported next to every headline metric.
- Results are broken down by year, strategy, time of day and market regime. An edge that exists in one year only is not an edge.
- The 12-month holdout is evaluated once. Its file is read-only and an access log records every read.
- Any experiment that fails its gate is written up in the experiment store, not deleted.

Costs, infrastructure and operations

The project can run at close to \$0 a month until live trading, and at most about \$100 a month afterwards. At \$500 positions and a plausible edge of 0.2R per trade, a fixed monthly cost matters, so paid data is bought only when an experiment shows it is needed.

Phase	Item	Cost
Research (weeks 1 to 12)	Alpaca free history and news, SEC EDGAR	\$0
Research	One-time labeling of historical headlines with Jev (\$0.042 per million input tokens, about 250 million tokens)	About \$10
Research, only if E0 finds missing delisted stocks	One month of a paid reference-data plan, then cancel	Tens of dollars, once
Shadow mode and pilot	Live data from scaled IEX polled over REST, if E1 passes	\$0 per month
Shadow mode and pilot, if E1 fails	Alpaca Algo Trader Plus, full real-time feed	\$99 per month
Shadow mode and pilot	Classifier calls, a few thousand headlines a day	A few dollars per month
Trading on XTB	Real shares in a USD account: no commission under €100,000 monthly turnover, then 0.2% (minimum €10)	Spread only, below the limit

Jev pricing and access come from early-access listings and should be rechecked at signup.

Your computer

- Recommended: 16 GB of memory, a 500 GB solid-state drive, a wired network connection. LightGBM trains on the processor, so no graphics card is needed.
- Disable sleep and automatic restarts during the session. The US regular session is normally 16:30 to 23:00 Bucharest time, with a few weeks each spring and autumn when US and European clock changes are out of step. The code uses New York time throughout.
- The scanner can rebuild its day state (VWAP, opening range, volume so far) from stored bars after a restart, so a crash costs minutes, not the day.
- Latency from Romania to US data servers is roughly 100 to 150 ms, which is irrelevant for signals you act on by hand.

Operations

- Secrets stay in a git-ignored .env file.
- Raw downloads can be re-fetched, so back up only live signals, the trade journal and the experiment store, weekly, to an external drive.
- A heartbeat check flags a stale data feed after 60 seconds, and the dashboard shows feed health beside the signals.
- A turnover tracker warns at 80% of the €100,000 monthly commission-free limit. Above it, a \$500 trade pays the €10 minimum, roughly 2% per side, which erases any edge.

Risks and open questions

The largest risk is that no setup has an edge after costs. The plan is built so that this is discovered in weeks 5 to 8 at near-zero cost, not after months of platform work.

Risk	Why it matters	Mitigation
No rule-based setup has an edge after costs	Retail intraday edges are thin and costs are large at \$500 positions	E3 is an early kill gate. Fall back to other horizons or setups before building more.

Risk	Why it matters	Mitigation
Backtest overfitting	Many thresholds, many trials, flattering results	Trial counting, deflated Sharpe, robustness grid (E4), locked holdout
Missing delisted stocks in the history	Survivorship bias inflates results	E0 audit. One month of paid reference data if needed.
Free IEX volume fails near the open	RVOL drives strategy A and the liquidity filter	E1 decides between paid data, later start, or price-only features at the open
News arrives too late to trade	Catalyst moves are often priced in within seconds to minutes	E10 measures the lag. The EDGAR dilution veto keeps its value even if speed does not.
Classifier vendor risk	Jev is closed, three weeks old and on a waitlist	Interface plus a fallback model, chosen by the E9 bake-off
Vendor terms change	Alpaca's news API launched as a beta with a possible future price	Keep a local archive of everything downloaded. Accessors isolate the vendor.
Execution gap	Hesitation and slippage on manual entries	Trade journal and E14 measure it directly
XTB specifics	Tradable instruments, spreads and pre-market access may differ from assumptions	E2 and the open questions below
Effort	Roughly 20 weeks part-time before the pilot starts	Phase gates allow stopping early with little sunk cost

Open questions

- Which of the roughly 600 universe stocks can you actually buy as real shares on XTB, and are fractional shares available for them? The universe must be intersected with XTB's tradable list, which you can export from xStation.
- Does XTB allow trading US stocks in pre-market, or only from the regular open?

- Can a Romanian resident open an Alpaca account for market data, and do the free-tier terms allow this use? Alpaca's documentation lists separate US, EU, UK and other regional versions, so check which one applies to you.
- How is short-term trading profit taxed in Romania for real shares at XTB? Confirm with an accountant, since it can change net expectancy.
- How many hours a week can you commit? The roadmap assumes 10 to 15.
- What account size do you expect to reach if the pilot succeeds? It decides when paid data and larger positions become worthwhile.

Next steps: the first two weeks

The first two weeks end with a small, audited pilot dataset and a repository whose layering is enforced by tests, so every later experiment starts from trusted data and a stable structure.

- ☐ Create the Alpaca account and API keys, set a SEC User-Agent with your email, and join the TypeSafe waitlist
- ☐ Export XTB's list of tradable US stocks from xStation and note whether fractional shares and pre-market trading are offered
- ☐ Create the repository skeleton with the five layers, the contracts folder and import-linter rules, and confirm an illegal import fails the build
- ☐ Build the utilities: clock, point-in-time view, rate limiter, config loader, structured logging
- ☐ Build the Alpaca bars accessor and reference accessor with recorded-HTTP tests, including the 15-minute end clamp
- ☐ Build the ingestion manager and run the pilot: all daily bars since 2016, minute bars (SIP and IEX) for 200 symbols over 3 years, news and EDGAR filings for the same symbols
- ☐ Run E0 on the pilot and write the data audit report
- ☐ Take a first quick look at E1 on the pilot data to size the volume-scaling question early
- ☐ Start hand-labeling headlines for E9, about 30 a day, in a spreadsheet
- ☐ Start the E2 log: XTB quotes beside Alpaca prices for 12 stocks, three times a day, for 5 days

Exit criteria: pilot data downloaded and audited, measured storage and download time for the full backfill, the architecture tests green, and a go or no-go on the full backfill.

Sources

Pages opened on 2026-10-05. Prices, limits and rules can change, so recheck them at signup.

- [XTB: API access discontinued 14 March 2025](#)
- [XTB: commissions and the 0.5% currency conversion fee](#)
- [Alpaca: market data plans, history since 2016, 15-minute limit, 30-symbol WebSocket](#)
- [Alpaca: historical news from Benzinga, 130+ articles a day](#)
- [Alpaca: end of the pattern day trader rule on 4 June 2026](#)
- [ThursdAI: Jev launch, pricing and waitlist access](#)

Not yet verified, and checked in the first sprint: SEC EDGAR submission field names, and whether Alpaca's asset list and history cover delisted symbols.